Alexandre Autran

Student in mathematics and machine learning

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Profile

I am a mathematics and machine learning student at ENS Rennes and ENS Paris-Saclay, interested in the mathematical foundations of machine learning and data science, with a particular focus on probabilistic methods, high-dimensional statistics, stochastic processes, and optimal transport.

Education

2024–2027 **Normalien – ENS Diploma in Mathematics**
Admitted to the ENS through a competitive entrance examination.
Four-year diploma completed in three years (accelerated track).

2026–2027 **Research Year in Machine Learning**, *École Normale Supérieure Paris-Saclay*, France
Final year of the ENS Diploma (research track).

2025–2027 **MSc in Mathematics**, *École Normale Supérieure de Rennes*, France
M2 Fundamental Mathematics. Highest honors (17/20 in the first semester).
M1 Fundamental Mathematics. Highest honors (17.6/20). Rank: 5/100+.

2024–2025 **BSc in Mathematics, Computer Science minor**, *École Normale Supérieure de Rennes*, France
L3 Fundamental Mathematics. High honors (15.4/20).

2022–2024 **BSc in Mathematics, Computer Science minor**, *University of Rennes*, France
L2 Mathematics. Highest honors (18/20). Rank: 3/100+.
L1 Mathematics. Highest honors (18.5/20). Rank: 1/100+.

Experiences [Full list here]

Sep 2026–Jan 2027 **Machine Learning Researcher**
High-Dimensional Density Analysis via Local Unimodality.
Supervised by Argyris Kalogeratos (Senior Researcher) at Centre Borelli, ENS Paris-Saclay, France.

May–Aug 2026 **M1 Research Internship**
Optical Modeling of VO_2-Based Thermochromic Oxide Nanocomposites.
Supervised by Nguyen Ngoc Duy (Prof.) and Amaury Baret (PhD Student) at the Univ. of Liège, Belgium.

Apr–Jun 2026 **Research Internship**
Numerical Simulation and Optimal Control of Spatial Lotka–Volterra Systems.
Supervised by Martin Gander (Prof.) and Bastien Chaudet-Dumas (Postdoctoral Researcher) at the University of Geneva, Switzerland.

Nov–Dec 2025 **M2 Research Seminar**
Point Processes and Stein's Method.
Supervised by Jean-Christophe Breton (Prof.) at the University of Rennes, France.

Jun–Aug 2025 **L3 Research Internship**
Lattice-Based Post-Quantum Cryptography.
Supervised by Pierre-Alain Fouque (Prof.) and Aurore Guillevic (Research Scientist) at Inria, France.

May–Jun 2025 **Research Internship**
Algebraic Counting of the Real Roots of Polynomials.
Supervised by Marie-Françoise Roy (Prof. Emeritus) at the University of Rennes, France.

Research [Full list here]

[1] **A. Autran**, Singular Tracking of Fractional Brownian Motion under Proportional Costs, preprint, 2026. Submitted.

[2] **A. Autran**, Improved Wasserstein Estimates for Jump-Diffusion Perturbations under Semigroup Smoothing, preprint, 2026. Submitted.

[3] **A. Autran**, Optimization with an Exact Discrete Adjoint for a Diffusive Lotka–Volterra Harvesting Problem, preprint, 2026.

[4] **A. Autran**, D. Munteanu, and J.-L. Autran, Multi-Dimensional Charge Diffusion–Collection Model in Semiconductor Devices Subjected to Single Ionizing Particles, *Applied Sciences*, vol. 16, no. 11, Art. no. 5551, 2026.

Other Activities [Full list here]

2024–2025 **ENS Oral Examinations**
Supervision of second-year BSc students in preparation for the oral entrance examinations in mathematics of the Écoles Normales Supérieures (Ulm, Paris-Saclay, Rennes, and Lyon).

2022–2025 **Teaching in Mathematics**
Tutoring for final-year high school students at Lycée Chateaubriand in Rennes, as well as first- and second-year BSc students at the University of Rennes.

2023–2024 **BSc Student Representative**
Representative on the council of the Mathematics Department of the University of Rennes.

Before 2022 **High School Competitions**
Participation in the Concours Général in Mathematics, Physics, and Philosophy, as well as the National High School Mathematics Olympiad, at Lycée Thiers in Marseille.

Computer Skills

Programming: Python (NumPy, pandas, scikit-learn, PyTorch), C/C++, Java, MATLAB, R.

Languages

French: native language. **English:** professional working proficiency.

Sports and Interests

Sports: running, cycling, strength training.
Interests: fantasy and science fiction literature; chess.

Academic Reference

François Bolley Email: francois.bolley@ens-rennes.fr.
Full Professor and Director of the Mathematics Department at ENS de Rennes.